

CS486 MOCK FINAL EXAMINATION - TEST 4

Based on the FINAL REVIEW structure; total: 10 points

Question 1 - ER Models

[1 pt]

An online marketplace stores **Sellers**, **Products**, **Customers**, **Orders**, and **OrderItems**. Seller IDs, Customer IDs, and Order IDs are unique. A product code is unique only within a seller. An order belongs to one customer and contains one or more products, with quantity and sale price recorded per item. The same product cannot appear twice in one order.

Draw the ER diagram and identify the key of Product and OrderItem after relational mapping.

Question 2 - Query Languages

[2 pt]

Schemas:

Seller(*SID*, *SName*), *Product*(*SID*, *PCode*, *PName*, *ListPrice*), *Orders*(*OID*, *CID*, *ODate*), *OrderItem*(*OID*, *PCode*, *Qty*, *SalePrice*)

Write:

- (a) RA to find sellers whose products have appeared in at least one order but for which no sold item was priced above its ListPrice.
- (b) TRC to find orders containing products from at least two different sellers.
- (c) SQL to find the top-selling product(s) by total quantity within each seller. Keep ties and include only products sold at least once.

Question 3 - Database Constraints

[1 pt]

Business rule: *For every OrderItem, Qty must be positive and SalePrice must be nonnegative. Also, an order cannot contain an item whose seller has been suspended before the order date.*

Assume *SellerStatus*(*SID*, *SuspendedFrom*). Give SQL CHECKs for row-local rules and a formal/trigger-based solution for the temporal cross-table rule.

Question 4 - Serialization & Concurrency

[2 pt]

Given

$$S = r_1(P) \ w_2(P) \ r_2(Q) \ w_3(Q) \ r_1(Q) \ w_1(R) \ r_3(R) \ w_2(R).$$

- (a) Construct the precedence graph and test conflict serializability.
- (b) Suppose T_1 reads the value of Q written by T_3 , and commit order is c_1, c_3, c_2 . Is the schedule recoverable? Is it cascadeless?
- (c) Explain how wait-die or wound-wait can prevent deadlock when transactions request locks in inconsistent orders.

Question 5 - Recovery**[1 pt]**

An immediate-update system uses UNDO/REDO. At crash:

```
<start ckpt (T1,T2,T3)>
<T1, X, 5, 8>
<T2, Y, 9, 12>
<commit T2>
<T3, Z, 4, 7>
<end ckpt>
<T1, X, 8, 11>
<commit T1>
<T4, W, 1, 2>
--- crash ---
```

Identify winners/losers and the values/actions that must be REDO/UNDO. Assume no later log records.

Question 6 - Functional Dependencies & Normalization**[1 pt]**

For $R(A, B, C, D, E)$ with

$$F = \{A \rightarrow B, B \rightarrow C, AC \rightarrow D, D \rightarrow E\},$$

- Find a candidate key.
- Determine the highest normal form.
- Decompose to 3NF or BCNF using dependency-preserving steps where possible, and state whether your decomposition is lossless.

Question 7 - Concurrency Control Execution**[1 pt]**

Inventory $Q = 10$.

```
T1: r(Q); if Q>=7 then Q:=Q-7; w(Q)
T2: r(Q); if Q>=5 then Q:=Q-5; w(Q)
```

Show an interleaving causing both transactions to approve their sale even though only 10 units exist. Give the final stored value and explain why application-level decisions can still be inconsistent even when the final numeric value looks plausible.

Question 8 - Query Optimization**[1 pt]**

OrderItem has 5,000,000 tuples and 50 tuples/page (100,000 pages). There is a clustered B+ tree on $(SID, PCode)$ of height 4 and an unclustered hash index on OID. Estimate I/O for:

- all items for one seller expected to occupy 800 contiguous data pages;
- all items for one specific order expected to contain 8 tuples scattered across pages.

Choose B+ tree/range scan, hash, or full scan for each and explain.

– End of Mock Test –